Neev: AI Facade Screening Platform

3-Minute Video Walkthrough Production Script & Cue Sheet

Presenter: Devansh
Engine: Meta SAM 3.1 + U-Net Hybrid
Target Time: ~2m 50s (Under 3:00 Max)

STRUCTURE Act 1 (Face) → Act 2 (Screen) → Act 3 (Face)

Audio Pacing: ~135 Words / Min (~400 words total)

Resolution: 1080p (1920×1080, 100% Zoom)

Server: http://127.0.0.1:8000

SEGMENT	VISUAL SOURCE	TIME RANGE	KEY TECHNICAL DEMONSTRATION
Act 1: Intro	FULL-FACE VIDEO	0:00 – 0:25 (25s)	The $15B facade inspection hazard, manual rope limits, introducing Neev built with Meta AI.
Act 2: Part A	SCREEN CAPTURE	0:25 – 0:55 (30s)	Baseline SAM 3.1 control: detects windows/building as scene context with 0 false alarms ([LOOKS FINE]).
Act 2: Part B	SCREEN CAPTURE	0:55 – 1:20 (25s)	Explaining Standard Whole-Image vs Adaptive Multi-View Tiling (4× effective pixel resolution) & Hybrid model.
Act 2: Part C	SCREEN CAPTURE	1:20 – 1:55 (35s)	Screening weathered wall: SHA-256 legal hash, SAM 3.1 grounding + local U-Net edge recovery, IoU deduplication.
Act 2: Part D	SCREEN CAPTURE	1:55 – 2:30 (35s)	14 cracks delineated, zoom on crack#1, 2D physics (skeleton, P95 width, shear angle), canvas reset & opacity.
Act 3: Outro	FULL-FACE VIDEO	2:30 – 2:55 (25s)	Human-in-the-loop triage mandate, physical NDT testing disclaimer, final project sign-off.

CAMERA: FULL-FACE ACT 1: On-Camera Introduction

0:00 – 0:25 (25 sec)

DIRECTOR CUE & FRAMING:
Medium close-up (chest-up). Camera lens at eye level. Look straight into the lens. Warm, confident, professional opening delivery.

> *"Hi everyone, I'm Devansh.*
>
> *Inspecting high-rise building facades today is one of the most hazardous and expensive challenges in civil engineering. Traditional manual inspections require scaffolding, rope access, and hundreds of human hours — yet subtle hairline fractures and joint separations still get missed until catastrophic spalling occurs.*
>
> *To solve this, I built* **Neev** *— an AI-powered facade screening platform developed with Meta AI. Neev pairs Meta's SAM 3.1 foundation model with specialized edge-recovery networks to triage building facades from ordinary photos in seconds.*
>
> *Let's dive into the workstation and see how it works."*

SCREEN: BROWSER

ACT 2 · PART A: Baseline SAM 3.1 Control & Zero False Alarms

0:25 – 0:55 (30 sec)

ON-SCREEN ACTIONS:
Start on `#/new`. Drop `photo_Sample/crack_wall.jpeg` into dropzone. Select *Meta SAM 3.1 (Primary Production)* and click *Standard Whole-Image*. Click *Execute Facade Screening* →. Hover over `[LOOKS FINE] 0 visual candidate crack feature(s)` banner. Point to sidebar showing *Scene Context: building* and *Scene Context: window* (Informational).

"Before an AI model can reliably find structural defects, it must first prove it won't cry wolf on ordinary architectural objects.

Here on the New Inspection tab, we test Meta's Segment Anything Model 3.1 using a single whole-image pass.

Look at the result: Neev accurately segments the windows and structural building contours as scene context, but reports zero defect candidates. It doesn't mistake a window sill or architectural shadow for a concrete crack. This high specificity eliminates false alarms right from the baseline."

SCREEN: BROWSER

ACT 2 · PART B: Multi-View Tiling & Architecture Strategy

0:55 – 1:20 (25 sec)

ON-SCREEN ACTIONS:
Click *New Inspection* in navbar. Notice active photo is ready to rescan. Move cursor to *2. Screening Configuration*: point at *Meta SAM 3.1*, then hover over *SAM + U-Net Hybrid*. Point at *Standard Whole-Image*, then click *Adaptive Multi-View (Tiled 2×2)*.

"Now, why do we have two execution strategies?

When an entire building facade is compressed into a single standard pass, subtle hairline fractures lose pixel resolution and get washed out.

Neev solves this with Adaptive Multi-View Tiling. The platform slices the facade into overlapping high-resolution tiles, effectively quadrupling the effective pixel density.

And for weathered masonry where micro-cracks blend into stucco textures, we switch from pure SAM 3.1 to our SAM plus U-Net Hybrid architecture."

SCREEN: BROWSER

ACT 2 · PART C: Distressed Facade Run & Model Mechanics

1:20 – 1:55 (35 sec)

ON-SCREEN ACTIONS:
Select *SAM + U-Net Hybrid (High-Recall)* and *Adaptive Multi-View*. Click *Rescan Current Photo* →. Screen displays pulsing radar: *"Processing Morphological Layers... Retrieving Facade Record & Reconciling Evidence"*. Keep screen steady on radar loader while delivering explanation.

> *"Let's execute the hybrid screening.*
>
> *While the engine processes, here is what's happening behind the scenes: First, every photo receives an immutable SHA-256 cryptographic hash to guarantee legal auditability for engineering reports.*
>
> *Next, our hybrid engine combines Meta SAM 3.1's zero-shot foundation understanding with a local high-recall U-Net model. SAM provides macro structural scene awareness, while U-Net isolates micro-fractures along mortar joints.*
>
> *Finally, our spatial fusion engine resolves overlapping tile detections using spatial intersection-over-union, delivering a single unified, deduplicated facade map."*

SCREEN: BROWSER

ACT 2 · PART D: Screened Findings, Physical Diagnostics & Canvas Reset

1:55 – 2:30 (35 sec)

ON-SCREEN ACTIONS:
Inspection view loads. Point to `[NEEDS INSPECTION] 14 visual candidate crack feature(s)`. Show red overlays. Click `Candidate: crack#1` in right sidebar → canvas zooms to 134%. Point to *Clear Selection (Reset View)* button. Click it (or press *Esc*) → canvas smoothly resets to whole facade. Drag *Mask Opacity* slider from 70% to 0% and back.

> *"And here are the results: 14 distinct visual crack candidates delineated with pinpoint precision.*
>
> *The system immediately flags the facade as 'Needs Inspection'.*
>
> *Clicking on this major vertical fracture on the left masonry wall instantly zooms in on the distress zone. Behind every finding, Neev computes deterministic 2D physical morphology: Zhang-Suen skeleton path length, orthogonal distance width profiling, and in-plane shear orientation.*
>
> *Inspectors can adjust mask opacity to visually verify the bare masonry, and simply hit 'Clear Selection' or tap Escape to restore the full facade view instantly."*

CAMERA: FULL-FACE ACT 3: On-Camera Conclusion

2:30 – 2:55 (25 sec)

DIRECTOR CUE & FRAMING:
Cut back to full-face camera. Same framing and lighting. Warm, grounded, authoritative engineering summary and sign-off.

> *"Crucially, Neev is designed as an intelligent triage tool — not a replacement for certified structural engineers. As highlighted in our platform disclaimers, visual 2D screening directs licensed inspectors to the exact zones requiring physical, non-destructive testing.*
>
> *By combining Meta SAM 3.1's vision intelligence with local high-recall edge recovery, Neev transforms hours of manual inspection into minutes of objective, auditable evidence.*
>
> *Thank you for watching Neev — built by Devansh with Meta AI."*

Production & Screen Recording Tips:

- **Display:** Record browser at 1920×1080 (1080p), zoom 100% or 110%. Hide bookmarks bar.
- **Mouse Movement:** Move smoothly and deliberately. Hover steadily on what you are talking about.
- **Audio Splicing:** If you make a mistake while speaking, pause 2 seconds and re-say that specific sentence; you can easily cut and align the audio in your editor.